

CLIMATEBERT: A Pretrained Language Model for Climate-Related Text

Bridging the Gap Between General NLP and Climate Science

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Paper Summary & Analysis

Overview of Presentation

- 1 The Problem with General AI
- 2 Methodology: How it was Built
- 3 Evaluation: Does it work?
- 4 Environmental Impact
- 5 Conclusion

The Context: NLP in Climate Science

The Rise of NLP

- Researchers increasingly use Natural Language Processing (NLP) to analyze climate disclosures, news, and scientific papers.
- Old methods used "Bag-of-Words" (counting keywords), which ignores context.
- Modern methods use **Transformers** (like BERT) to understand context.

The Problem

Standard models (BERT, RoBERTa) are trained on *general* text (Wikipedia, Books). They struggle with **niche** climate language.

Example: To a general model, "environment" might mean a software setting. To a climate model, it means the ecosystem.

The Solution: CLIMATEBERT

What is CLIMATEBERT?

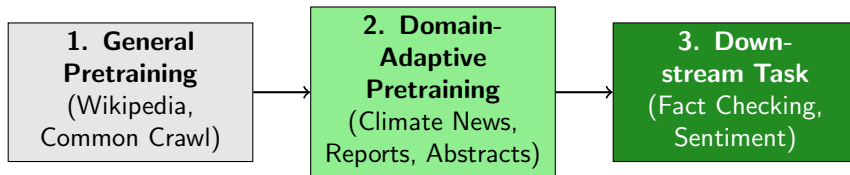
A transformer-based language model further pretrained on over **2 million paragraphs** of climate-related text.

Key Achievements:

- **48% improvement** on masked language modeling objectives compared to the base model.
- Reduces error rates by **3.57% to 35.71%** on downstream tasks.
- Sets a new State-of-the-Art (SotA) for climate fact-checking.

The Training Pipeline

The authors used a technique called **Domain-Adaptive Pretraining**.



- **Base Model:** DistilRoBERTa (Chosen for efficiency and lower carbon footprint).
- **Goal:** Teach the model the specific "language" of climate science before asking it to solve specific problems.

The Data: Where did the text come from?

The model was trained on a custom corpus called **CORP** containing 2 million paragraphs.

Dataset	Source	Content Type
News	Refinitiv & Web	Articles on floods, droughts, politics
Abstracts	Web of Science	Scientific summaries (2000-2019)
Reports	Corporate Disclosures	Sustainability reports (600+ companies)
Total	2,046,523 Paragraphs	

Table: Composition of the training corpus

Note: Full research articles were excluded as they were "too specific." Twitter was excluded as "too noisy."

The Concept: Tokenization

AI doesn't read words; it reads "tokens" (parts of words). If a word isn't in its vocabulary, it breaks it into nonsense chunks.

The Modification:

- The authors analyzed the climate corpus vs. general corpus.
- They added **235 new tokens** specifically for climate concepts.
- **Examples of added tokens:** "CO2", "emissions", "renewable", "fossil", "adaptation".
- **Result:** The model processes these critical terms as whole concepts, not fragments.

Sample Selection Strategies

Does using *all* data make the model better? Not always. The authors experimented with selecting specific data to pretrain on.

- ➊ **Full-Select (CLIMATEBERT-F):** Used all 2M paragraphs.
- ➋ **Sim-Select (CLIMATEBERT-S):** Selected the 70% of data most *similar* to the target tasks (using Euclidean distance).
- ➌ **Div-Select (CLIMATEBERT-D):** Selected the 70% of data most *diverse* (using Entropy) to force the model to learn new things.
- ➍ **Combined (CLIMATEBERT-D+S):** A mix of diversity and similarity.

Finding: Generally, simple Domain-Adaptive Pretraining (using the full corpus or similar corpus) yielded the best loss reduction.

Downstream Task 1: Text Classification

Objective: Decide if a paragraph from a corporate report is climate-related or not.

- **Inputs:** Paragraphs from annual reports.
- **Labels:** Yes / No.
- **Annotation:** Hand-labeled by experts.

Results

- Base Model (DistilRoBERTa) F1 Score: 0.986
- **CLIMATEBERT-S F1 Score: 0.991**
- **Impact:** Error rate reduced by 35.71%. The model is significantly better at distinguishing "greenwashing" fluff from actual climate relevance.

Downstream Task 2: Sentiment Analysis

Objective: Determine the sentiment of a climate disclosure.

- **Labels:**

- ① **Risk:** (Negative) e.g., "Floods will damage our factory."
- ② **Opportunity:** (Positive) e.g., "We can sell more solar panels."
- ③ **Neutral:** Just stating facts.

Results

- Base Model Loss: 0.150
- **CLIMATEBERT-F Loss: 0.139** (7.33% lower loss)
- **Why it matters:** Helps investors understand if companies view climate change as a threat or a business chance.

Downstream Task 3: Fact-Checking

Objective: Verify climate claims against evidence.

- **Dataset:** CLIMATE-FEVER (Real-world climate claims).
- **Task:** Given a Claim and Evidence, classify as **Support** or **Refute**.

Claim (Refute)	Evidence
"Global warming has been disproven."	"97% consensus on human-caused global warming..."

New State-of-the-Art

CLIMATEBERT achieved an F1 score of **0.757**, beating the previous best score of 0.718. It understands the nuance of scientific claims better than general models.

Green AI: The Carbon Footprint

Training AI consumes massive amounts of electricity. The authors were conscious of this.

Strategies to Reduce Emissions:

- Used **DistilRoBERTa** (smaller model) instead of full BERT.
- Limited hyperparameter tuning (random guessing of settings).
- Used energy-efficient NVIDIA RTX A5000 GPUs.

Total Footprint

115.15 kg
CO₂

(Equivalent to driving a car $\approx$ 450 km)

Offsetting: The authors donated to *atmosfair* to offset these emissions, supporting renewable energy projects.

Summary Takeaways

- **The Gap:** General NLP models fail at the specific, technical, and ambiguous language of climate change.
- **The Tool:** CLIMATEBERT fills this gap via Domain-Adaptive Pretraining and vocabulary augmentation.
- **The Performance:** consistently outperforms standard models in:
 - Identifying climate text.
 - Detecting risks vs. opportunities.
 - Fact-checking misinformation.
- **Availability:** The model is open-source (HuggingFace) to encourage further research.

"We see high potential of NLP to support action against climate change."

Questions?

Resource: Model available at huggingface.co/climatebert